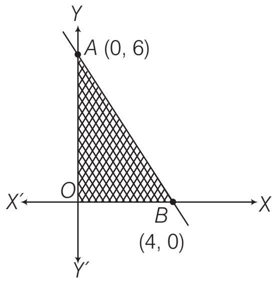
**73.** The shaded region, including the boundary in the given graph, is exactly represented by




(a) $3x + 2y \le 12, x < 0, y \ge 0$ (b) $3x + 2y \le 12, x \ge 0, y \ge 0$
(c) $3x + 2y < 12, x \ge 0, y \ge 0$ (d) $3x + 2y > 12, x \ge 0, y \ge 0$

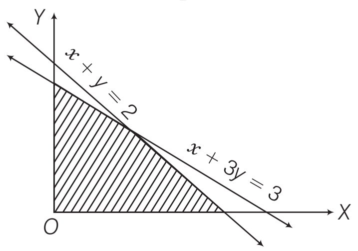
**74.** When is the expression $x^2 + 3x - 10$ positive only?
(a) $x \le -5$ (b) $x \ge 2$ (c) $-5 < x < 2$ (d) $x < -5$ or $x > 2$

**75.** The solution set for the quadratic inequation $x^2 - 5x + 6 \ge 0$ is
(a) $[-\infty, 2] \cup [3, \infty]$ (b) $[-\infty, 2] \cup [3, \infty]$
(c) $[-\infty, 2] \cup [3, \infty]$ (d) $[2, 3]$

**76.** All real values of $x$ for which $\dfrac{x-2}{3x+1} < \dfrac{x-3}{3x-2}$ are
(a) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (b) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (c) $\left[\dfrac{-1}{3}, \dfrac{2}{3}\right]$ (d) $R$

**77.** If $x + 2y \le 3, x > 0$ and $y > 0$, then one of the solution is
(a) $x = -1, y = 2$ (b) $x = 2, y = 1$
(c) $x = 1, y = 1$ (d) $x = 0, y = 0$

**78.** The shaded region in the given figure is the solution set of the inequalities




(a) $x + y \le 2, x + 3y \ge 3, x \ge 0, y \ge 0$
(b) $x + y \ge 2, x + 3y \ge 3, x \ge 0, y \ge 0$
(c) $x + y \ge 2, x + 3y \le 3, x \ge 0, y \ge 0$
(d) $x + y \le 2, x + 3y \le 3, x \ge 0, y \ge 0$

**79.** If $a + b = 2m^2$, $b + c = 6m$, $a + c = 2$, where $m$ is a real number and $a \le b \le c$, then which one of the following is correct?
(a) $0 \le m \le 1/2$ (b) $-1 \le m \le 0$
(c) $1/3 \le m \le 1$ (d) $1 < m \le 2$

**80.** If $\alpha$ and $\beta$ are the roots of the equation $(x^2 - 3x + 2 = 0)$, then which of the following equation has the roots $(\alpha + 1)$ and $(\beta + 1)$?
I. $x^2 + 5x + 6 = 0$ II. $x^2 - 5x - 6 = 0$

*Select the answer using the codes given below.*
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

**81.** Consider the following statements
I. Every quadratic equation has atleast one real root.
II. A quadratic equation with integral coefficients has integral roots.
III. If the coefficient of $x^2$ and the constant term of a quadratic equation have opposite signs, then the quadratic equation has real roots.

*Which of the above statements is/are correct?*
(a) Only I and II (b) Only II and III
(c) Only III (d) All of these

**82.** Which of the following are quadratic equations?
I. $x^2 + \dfrac{1}{x^2} = 2$ II. $x + \dfrac{3}{x} = x^2$
III. $2x^2 - x + 2 = x^2 + 4x - 4$ IV. $x^3 + 6x^2 + 2x - 1 = 0$

*Select the correct answer using the codes given below.*
(a) I, III are quadratic (b) II, III and IV are quadratic
(c) Only III is quadratic (d) None of these

**83.** Consider the following statements
I. $x = 1$ is a root of $3x^2 - 2x - 1 = 0$.
II. $x = -2\sqrt{2}$ is a root of $x^2 + \sqrt{2}x - 4 = 0$.
III. $x^2 + x + 1 = 0$; $x = 1, x = -1$.
IV. $9x^2 - 3x - 2 = 0$; $x = -\dfrac{1}{3}, x = \dfrac{2}{3}$.

*Which of the equation(s) given above is/are correct?*
(a) I and II (b) I, II and IV
(c) III and IV (d) Only IV

**Directions** (Q. Nos. 84-86) *A ball is thrown upwards from a rooftop, 80 m above the ground. It will reach a maximum vertical height and then fall back to the ground. The height of the ball from the ground at time t is h which is given by,* $h = -16t^2 + 64t + 80$ *Now, answer the following questions based on the above information.*

**84.** What is the height reached by the ball after 1 sec?
(a) 150 m (b) 128 m
(c) 64 m (d) None of these

**85.** What is the maximum height reached by the ball?
(a) 110 m (b) 132 m
(c) 144 m (d) cannot be determined

**86.** How long will it take before hitting the ground?
(a) 5 sec (b) 6 sec
(c) 3 sec (d) cannot say

## PREVIOUS YEARS' QUESTIONS

**87.** If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are real and less than 3, then which one of the following is correct? **2012 I**
(a) $a < 2$ (b) $2 < a < 3$ (c) $3 < a < 4$ (d) $a > 4$

**88.** If one of the roots of quadratic equation $7x^2 - 50x + k = 0$ is 7, then what is the value of $k$? **2012 I**
(a) 7 (b) 1 (c) 50/7 (d) 7/50

**89.** Two students $A$ and $B$ solve an equation of the form $x^2 + px + q = 0$. $A$ starts with a wrong value of $p$ and obtains the roots as 2 and 6. $B$ starts with a wrong value of $q$ and gets the roots as 2 and $-9$. What are the correct roots of the equation? **2012 II**
(a) 3 and $-4$ (b) $-3$ and $-4$ (c) $-3$ and 4 (d) 3 and 4

**90.** If one of the roots of the equation $x^2 - bx + c = 0$ is the square of the other, then which of the following option is correct? **2013 I**
(a) $b^3 = 3bc + c^2 + c$ (b) $c^3 = 3bc + b^2 + b$
(c) $3bc = c^3 + b^2 + b$ (d) $3bc = c^3 + b^3 + b^2$

**91.** The difference of the roots of the equation $2x^2 - 11x + 5 = 0$ is **2013 I**
(a) 4.5 (b) 4 (c) 3.5 (d) 3

**92.** The sum of the squares of two numbers is 97 and the squares of their difference is 25. The product of the two numbers is **2013 I**
(a) 45 (b) 36 (c) 54 (d) 63

**93.** If $x + \frac{1}{x} = 2$, then what is value of $x - \frac{1}{x}$? **2013 I**
(a) 0 (b) 1 (c) 2 (d) $-2$

**94.** There are some benches in a classroom having the number of rows 4 more than the number of columns. If each bench is seated with 5 students, there are two seats vacant in a class of 158 students. The number of rows is **2013 I**
(a) 4 (b) 8 (c) 6 (d) 10

**95.** Which one of the following is factor of
$$x^2 + \frac{1}{x^2} + 8\left(x + \frac{1}{x}\right) + 14?$$ **2013 II**
(a) $x + \frac{1}{x} + 1$ (b) $x + \frac{1}{x} + 3$
(c) $x + \frac{1}{x} + 6$ (d) $x + \frac{1}{x} + 7$

**96.** If $\alpha$ and $\beta$ are the roots of the equation $x^2 - x - 1 = 0$, then what is $\frac{\alpha^2 + \beta^2}{(\alpha^2 - \beta^2)(\alpha - \beta)}$ equal to? **2013 II**
(a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) $\frac{4}{5}$ (d) None of these

**97.** If $x^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \cdots \infty}}}$, then what is one of the values of $x$ equal to? **2013 II**
(a) 6 (b) 5 (c) 4 (d) 3

**98.** Consider the following statements in respect of the quadratic equation $ax^2 + bx + b = 0$, where $a \neq 0$.
I. The product of the roots is equal to the sum of the roots.
II. The roots of the equation are always unequal and real.

*Which of the above statements is/are correct?* **2014 I**
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

**99.** If the roots of the equation $Ax^2 + Bx + C = 0$ are $-1$ and 1, then which one of the following is correct? **2014 I**
(a) $A$ and $C$ are both zero
(b) $A$ and $B$ are both positive
(c) $A$ and $C$ are both negative
(d) $A$ and $C$ are of opposite sign

**100.** If the roots of the equation $(a^2 - bc)x^2 + 2(b^2 - ac)x + (c^2 - ab) = 0$ are equal, where $b \neq 0$, then which one of the following is correct? **2014 I**
(a) $a + b + c = abc$ (b) $a^2 + b^2 + c^2 = 0$
(c) $a^3 + b^3 + c^3 = 0$ (d) $a^3 + b^3 + c^3 = 3abc$

**101.** If $m$ and $n$ are the roots of the equation $ax^2 + bx + c = 0$, then the equation whose roots are $(m^2 + 1)/m$ and $(n^2 + 1)/n$ is **2014 I**
(a) $acx^2 + (ab + bc)x + b^2 + (a - c)^2 = 0$
(b) $acx^2 + (ab - bc)x + b^2 + (a - c)^2 = 0$
(c) $acx^2 + (ab - bc)x + b^2 - (a - c)^2 = 0$
(d) $acx^2 + (ab + bc)x + b^2 - (a - c)^2 = 0$

**102.** In solving a problem, one student makes a mistake in the coefficient of the first degree term and obtains $-9$ and $-1$ for the roots. Another student makes a mistake in the constant term of the equation and obtains 8 and 2 for the roots. The correct equation was **2014 I**
(a) $x^2 + 10x + 9 = 0$ (b) $x^2 - 10x + 16 = 0$
(c) $x^2 - 10x + 9 = 0$ (d) None of these

**103.** If $m$ and $n$ $(m > n)$ are the roots of the equation $7(x + 2a)^2 + 3a^2 = 5a(7x + 23a)$, where $a > 0$, then what is $3m - n$ equal to? **2014 II**
(a) $12a$ (b) $14a$ (c) $15a$ (d) $18a$

**104.** If one of the roots of the equation $px^2 + qx + r = 0$ is three times the other, then which one of the following relations is correct? **2014 II**
(a) $3q^2 = 16pr$ (b) $q^2 = 24pr$
(c) $p = q + r$ (d) $p + q + r = 1$

**105.** If $m$ and $n$ are the roots of the equation $x^2 + ax + b = 0$ and $m^2$, $n^2$ are the roots of the equation $x^2 - cx + d = 0$, then which of the following is/are correct?

I. $2b - a^2 = c$ II. $b^2 = d$

*Select the correct answer using the codes given below.* **2014 II**

(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

**106.** The sign of the quadratic polynomial $ax^2 + bx + c$ is always positive, if **2015 I**

(a) $a$ is positive and $b^2 - 4ac \le 0$
(b) $a$ is positive and $b^2 - 4ac \ge 0$
(c) $a$ can be any real number and $b^2 - 4ac \le 0$
(d) $a$ can be any real number and $b^2 - 4ac \ge 0$

**107.** The number of values of $x$ satisfying $x + \frac{100}{x} > 50$, where $x$ is a natural number less than or equal to 100, is **2015 II**

(a) 51 (b) 53 (c) 55 (d) 57

**108.** If the roots of the quadratic equation $x^2 - 4x - \log_{10} N = 0$ are all real, then the minimum value of $N$ is **2015 II**

(a) $\frac{1}{100}$ (b) $\frac{1}{1000}$ (c) $\frac{1}{10000}$ (d) 10000

**109.** What is $\sqrt{4 + \sqrt{4 - \sqrt{4 +}}}\ldots$ equal to? **2015 II**

(a) 3 (b) $\frac{\sqrt{13} - 1}{2}$ (c) $\frac{\sqrt{13} + 1}{2}$ (d) 0

**110.** If $k = x - y + 2z$, where $-2 \le x \le 1$ and $-1 \le y \le 2$ and $3 \le z \le 6$, then which one of the following is correct? **2015 II**

(a) $0 \le k \le 9$ (b) $5 \le k \le 11$ (c) $2 \le k \le 14$ (d) $2 \le k \le 11$

**111.** If the sum of the roots of $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then which one of the following relations is correct? **2016 I**

(a) $ab^2 + bc^2 = 2a^2c$ (b) $ac^2 + bc^2 = 2b^2a$
(c) $ab^2 + bc^2 = a^2c$ (d) $a^2 + b^2 + c^2 = 1$

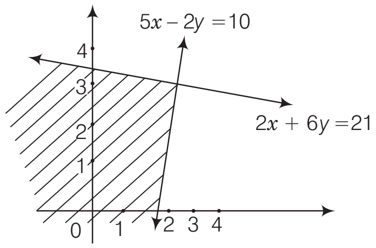
**112.** If the equations $x^2 - px + q = 0$ and $x^2 + qx - p = 0$ have a common root, then which one of the following is correct? **2016 I**

(a) $p - q = 0$ (b) $p + q - 2 = 0$
(c) $p + q - 1 = 0$ (d) $p - q - 1 = 0$

**113.** Under what condition on $p$ and $q$, one of the roots of the equation $x^2 + px + q = 0$ is the square of the other? **2016 I**

(a) $1 + q + q^2 = 3pq$ (b) $1 + p + p^2 = 3pq$
(c) $p^3 + q + q^2 = 3pq$ (d) $q^3 + p + p^2 = 3pq$

**114.**




The linear inequations, for which the shaded area in the figure given above is the solution set, are **2016 I**

(a) $2x + 6y \le 21$, $5x - 2y \le 10$
(b) $2x + 6y \le 21$, $5x - 2y \ge 10$
(c) $2x + 6y \ge 21$, $5x - 2y \le 10$
(d) $2x + 6y \ge 21$, $5x - 2y \ge 10$

**115.** Let $p$ and $q$ be non-zero integers. Consider the polynomial $A(x) = x^2 + px + q$. It is given that $(x - m)$ and $(x - km)$ are simple factors of $A(x)$, where $m$ is a non-zero integer and $k$ is a positive integer, $k \ge 2$. Which one of the following is correct? **2016 I**

(a) $(k + 1)^2 p^2 = kq$ (b) $(k + 1)^2 q = kp^2$
(c) $k^2 q = (k + 1)p^2$ (d) $k^2 p^2 = (k + 1)^2 q$

**116.** Which of the points $P(5, -1)$, $Q(3, -2)$ and $R(1, 1)$ lie in the solution of the system of inequations $x + y \le 4$ and $x - y \ge 2$? **2016 I**

(a) $Q$ and $R$ (b) $P$ and $R$
(c) $P$ and $Q$ (d) $P$, $Q$ and $R$

**117.** If $x = \frac{\sqrt{a + 2b} + \sqrt{a - 2b}}{\sqrt{a + 2b} - \sqrt{a - 2b}}$, then $bx^2 - ax + b$ is equal to (given that, $b \ne 0$) **2016 I**

(a) 0 (b) 1 (c) $ab$ (d) $2ab$

**118.** If $\sqrt{3x^2 - 7x - 30} - \sqrt{2x^2 - 7x - 5} = x - 5$ has $\alpha$ and $\beta$ as its roots, then the value of $\alpha\beta$ is **2016 I**

(a) – 15 (b) – 5 (c) 0 (d) 5

**119.** If the roots of the equation $lx^2 + mx + m = 0$ are in the ratio $p : q$, then $\sqrt{\frac{p}{q}} + \sqrt{\frac{q}{p}} + \sqrt{\frac{m}{l}}$ is equal to **2016 I**

(a) 0 (b) 1 (c) 2 (d) 3

**120.** The solution of the inequation $1 + \frac{1}{x} - \frac{1}{x^2} \ge 0$ is (given that, $x \ne 0$) **2016 I**

(a) $x > 0$ (b) $x < 0$
(c) $\frac{-1 - \sqrt{5}}{5} \le x \le \frac{-1 + \sqrt{5}}{2}$ (d) $x \le \frac{-1 - \sqrt{5}}{2}$ or $x \ge \frac{-1 + \sqrt{5}}{2}$

## ANSWERS

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | *b* | 2 | *c* | 3 | *a* | 4 | *b* | 5 | *c* | 6 | *a* | 7 | *c* | 8 | *d* | 9 | *d* | 10 | *b* |
| 11 | *b* | 12 | *a* | 13 | *b* | 14 | *b* | 15 | *c* | 16 | *b* | 17 | *c* | 18 | *b* | 19 | *d* | 20 | *c* |
| 21 | *d* | 22 | *b* | 23 | *a* | 24 | *b* | 25 | *a* | 26 | *a* | 27 | *c* | 28 | *d* | 29 | *d* | 30 | *c* |
| 31 | *c* | 32 | *b* | 33 | *b* | 34 | *a* | 35 | *c* | 36 | *b* | 37 | *c* | 38 | *b* | 39 | *b* | 40 | *b* |
| 41 | *b* | 42 | *d* | 43 | *d* | 44 | *d* | 45 | *a* | 46 | *c* | 47 | *b* | 48 | *a* | 49 | *c* | 50 | *a* |
| 51 | *c* | 52 | *d* | 53 | *c* | 54 | *c* | 55 | *b* | 56 | *d* | 57 | *d* | 58 | *c* | 59 | *a* | 60 | *a* |
| 61 | *c* | 62 | *a* | 63 | *b* | 64 | *c* | 65 | *a* | 66 | *b* | 67 | *b* | 68 | *c* | 69 | *b* | 70 | *c* |
| 71 | *b* | 72 | *a* | 73 | *b* | 74 | *d* | 75 | *a* | 76 | *b* | 77 | *c* | 78 | *d* | 79 | *c* | 80 | *d* |
| 81 | *c* | 82 | *c* | 83 | *b* | 84 | *b* | 85 | *c* | 86 | *a* | 87 | *a* | 88 | *a* | 89 | *b* | 90 | *a* |
| 91 | *a* | 92 | *b* | 93 | *a* | 94 | *b* | 95 | *c* | 96 | *b* | 97 | *d* | 98 | *d* | 99 | *d* | 100 | *d* |
| 101 | *a* | 102 | *c* | 103 | *c* | 104 | *a* | 105 | *b* | 106 | *a* | 107 | *c* | 108 | *c* | 109 | *c* | 110 | *c* |
| 111 | *a* | 112 | *d* | 113 | *c* | 114 | *a* | 115 | *b* | 116 | *c* | 117 | *a* | 118 | *a* | 119 | *a* | 120 | *d* |

# HINTS AND SOLUTIONS

**1.** *(b)* An equation of $n$ degree has $n$ roots. So, quadratic equation has two roots.

**2.** *(c)* $a^2b^2x^2 - a^2x - b^2x + 1 = 0$

$\Rightarrow \quad a^2x(b^2x - 1) - 1(b^2x - 1) = 0$

$\Rightarrow \quad (b^2x - 1)(a^2x - 1) = 0$

So, either $a^2x - 1 = 0 \Rightarrow a^2x = 1$

or $\quad b^2x - 1 = 0$ or $b^2x = 1$

$\Rightarrow \quad x = 1/a^2$ or $x = 1/b^2$

**3.** *(a)* Given, $ax^2 - 2\sqrt{5}x + 4 = 0$ has equal roots.

$\therefore$ Discriminant

$= (-2\sqrt{5})^2 - 4(a)4 = 0 \quad [\because D = B^2 - 4AC]$

$\Rightarrow \quad 20 - 16a = 0 \Rightarrow a = 5/4$

**4.** *(b)* Given, $3x^2 = 8x + (2k + 1)$

or $\quad 3x^2 - 8x - (2k + 1) = 0$

Let first and second root be $\alpha$ and $7\alpha$, respectively.

$\therefore$ Sum of roots $= \alpha + 7\alpha = \dfrac{8}{3}$

$\Rightarrow \quad 8\alpha = \dfrac{8}{3} \Rightarrow \alpha = \dfrac{1}{3}$

So, roots are $\dfrac{1}{3}$ and $\dfrac{7}{3}$.

Also, product of roots $= \dfrac{1}{3} \times \dfrac{7}{3} = \dfrac{7}{9}$

$\Rightarrow \quad \dfrac{7}{9} = \dfrac{-(2k + 1)}{3}$

$\Rightarrow \quad \dfrac{7}{3} = -(2k + 1)$

$\Rightarrow \quad 7 = -6k - 3 \Rightarrow 10 = -6k$

$\Rightarrow \quad k = \dfrac{-5}{3}$

**5.** *(c)* Here, sum of roots,

$$S = \frac{4 + \sqrt{7}}{2} + \frac{4 - \sqrt{7}}{2} = 4$$

Product of roots,

$$P = \left(\frac{4 + \sqrt{7}}{2}\right)\left(\frac{4 - \sqrt{7}}{2}\right) = \frac{16 - 7}{4} = \frac{9}{4}$$

The required equation is

$$x^2 - Sx + P = 0$$

$$x^2 - 4x + \frac{9}{4} = 0$$

$\Rightarrow \quad 4x^2 - 16x + 9 = 0$

**6.** *(a)* Given, $x^2 - 8x + p = 0$

Sum of roots $\alpha + \beta = 8$ and product of roots $\alpha\beta = p$

$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad \ldots\text{(i)}$

$\Rightarrow \quad 40 = (8)^2 - 2(p) [\because \alpha^2 + \beta^2 = 40]$

$\Rightarrow \quad 40 - 64 = -2p$

$\Rightarrow \quad -24 = -2p \Rightarrow p = 12$

**7.** *(c)* Given, $x^2 - 5x + 6 = 0$

$$\text{Sum of roots} = -\frac{\text{Coefficient of } x}{\text{Coefficient of } x^2}$$

$\Rightarrow \quad \alpha + \beta = 5$

$$\text{Product of roots} = \frac{\text{Constant term}}{\text{Coefficient of } x^2}$$

$\Rightarrow \quad \alpha\beta = 6$

Now, $\quad (\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$= (5)^2 - 4 \times 6 = 1$

$\Rightarrow \quad \alpha - \beta = \pm 1$ and $\alpha + \beta = 5$

$\Rightarrow \quad \alpha^2 - \beta^2 = (\alpha + \beta)(\alpha - \beta)$

$= 5(\pm 1)$

$\Rightarrow \quad \alpha^2 - \beta^2 = \pm 5$

**8.** *(d)* Here, $\alpha + \beta = -b/a$ and $\alpha\beta = c/a$

Now, roots of required equation are $\dfrac{1}{\alpha}, \dfrac{1}{\beta}$.

$$\Rightarrow S = \frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{-b/a}{c/a} = \frac{-b}{c}$$

$$P = \frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{c/a} = \frac{a}{c}$$

$\therefore$ Required quadratic equation is

$$x^2 - Sx + P = 0$$

$$\Rightarrow \quad x^2 - \left(\frac{-b}{c}\right)x + \frac{a}{c} = 0$$

$\Rightarrow \quad cx^2 + bx + a = 0$

**9.** *(d)* Given, $\alpha + \beta = 24$ and $\alpha - \beta = 8$

On solving, we get $\alpha = 16$ and $\beta = 8$

$\therefore$ Sum of roots $= \alpha + \beta = 24$

and product of roots $= 16 \times 8 = 128$

So, required equation is

$$x^2 - 24x + 128 = 0.$$

**10.** *(b)* Let $\alpha$ and $\beta$ be the roots of the equation $ax^2 + bx + c = 0$.

Then, $\quad \alpha + \beta = \dfrac{-b}{a}$ and $\alpha\beta = \dfrac{c}{a}$

Now, $3\alpha + 3\beta = \dfrac{-3b}{a}$ and $3\alpha \cdot 3\beta = \dfrac{9c}{a}$

$\therefore$ Required equation $= x^2 -$ (Sum of roots) $x$ + Product of roots

$$\Rightarrow \quad x^2 - \left(\frac{-3b}{a}\right)x + \frac{9c}{a} = 0$$

$\Rightarrow \quad ax^2 + 3bx + 9c = 0$

**11.** *(b)* Given equation is

$$x^{2/3} + x^{1/3} - 2 = 0$$

$\Rightarrow \quad (x^{1/3})^2 + x^{1/3} - 2 = 0$

Let $\quad x^{1/3} = x$

$\Rightarrow \quad x^2 + x - 2 = 0,$

It is a quadratic equation in $x$.

$\therefore$ Discriminant of $x^2 + x - 2 = 0$ is

$B^2 - 4\,AC = 1^2 - 4(1)(-2) = 9 > 0$

Hence, two real values of $x$ satisfy the given equation.

**12.** *(a)* Here, $\alpha + \beta = \frac{4}{2} = 2, \alpha\beta = \frac{1}{2}$

Now, $\frac{1}{\alpha + 2\beta} + \frac{1}{\beta + 2\alpha}$

$= \frac{\beta + 2\alpha + \alpha + 2\beta}{(\alpha + 2\beta)(\beta + 2\alpha)}$

$= \frac{3\alpha + 3\beta}{\alpha\beta + 2\alpha^2 + 2\beta^2 + 4\alpha\beta}$

$= \frac{3(\alpha + \beta)}{2(\alpha + \beta)^2 + \alpha\beta} = \frac{3(2)}{2(2)^2 + \frac{1}{2}} = \frac{12}{17}$

**13.** *(b)* Given equation is $2^{x+2} + 2^{-x} = 5$

$\Rightarrow \quad 2^x 2^2 + 2^{-x} = 5 \Rightarrow 4 \cdot 2^x + \frac{1}{2^x} = 5$

Put $\quad 2^x = y,\ 4y + \frac{1}{y} = 5$

$\Rightarrow \quad 4y^2 + 1 = 5y \Rightarrow 4y^2 - 5y + 1 = 0$

$\Rightarrow \quad (y - 1)(4y - 1) = 0$

$\Rightarrow \quad y - 1 = 0 \quad \text{or} \quad 4y - 1 = 0$

$\Rightarrow \quad y = 1 \quad \text{or} \quad 4y = 1$

$\Rightarrow \quad y = 1, \frac{1}{4}$

By condition,

| | |
|---|---|
| $2^x = y$ | $2^x = y$ |
| $\Rightarrow \quad 2^x = 1$ | $\Rightarrow \quad 2^x = \frac{1}{4}$ |
| $\Rightarrow \quad 2^x = 2^0$ | $\Rightarrow \quad 2^x = \frac{1}{2^2} = 2^{-2}$ |
| $\therefore \quad x = 0$ | $\therefore \quad x = -2$ |

Hence, the roots of the equation are 0 and $-2$.

**14.** *(b)* Since, $\alpha$ and $\beta$ are roots of the equation $x^2 + p = 0$.

$\therefore \quad \alpha + \beta = 0 \quad \text{and} \quad \alpha\beta = p$

So, $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = 0$

and $\frac{1}{\alpha} \cdot \frac{1}{\beta} = \frac{1}{\alpha\beta} = \frac{1}{p}$

$\therefore$ Required equation $x^2 - 0 \cdot x + \frac{1}{p} = 0$

$\Rightarrow \quad px^2 + 1 = 0$

**15.** *(c)* The equation is $x^2 + px + q = 0$.

Sum of roots $= -p = 1 + 2$

$\Rightarrow \quad p = -3$

Product of roots $= q = 1 \times 2 = 2$

$\therefore$ Equation $qx^2 - px + 1 = 0$ becomes

$$2x^2 - (-3)x + 1 = 0$$

$\Rightarrow \quad 2x^2 + 3x + 1 = 0$

$\Rightarrow \quad (2x + 1)(x + 1) = 0$

$\therefore \quad x = -\frac{1}{2}$ or $x = -1$

**16.** *(b)* Given, $2x^2 - 3x - 4 = 0$

For getting a reciprocal roots, we replace $x$ by $\frac{1}{x}$, we get

$$2\left(\frac{1}{x}\right)^2 - 3\left(\frac{1}{x}\right) - 4 = 0$$

$\Rightarrow \quad \frac{2}{x^2} - \frac{3}{x} - 4 = 0$

$\Rightarrow \quad -4x^2 - 3x + 2 = 0$

$\Rightarrow \quad 4x^2 + 3x - 2 = 0$

**17.** *(c)* As, $\sqrt{x + 4} = x - 2$

On squaring both sides, we get

$$(x + 4) = (x - 2)^2$$

$\Rightarrow \quad x + 4 = x^2 + 4 - 4x$

$\Rightarrow \quad x^2 - 5x = 0 \Rightarrow x = 0, x = 5$

But for $x = 0$, $\sqrt{0 + 4} = 0 - 2$

$\sqrt{4} \neq -2$

So, $x = 5$ is the only solution.

**18.** *(b)* Let roots of equation be $\alpha$ and $\frac{1}{\alpha}$.

$\therefore$ Product of roots

$= \alpha \times \frac{1}{\alpha} = \frac{\text{Constant term}}{\text{Coefficient of } x^2} = \frac{r}{p}$

$\Rightarrow \quad 1 = \frac{r}{p} \quad \Rightarrow \quad r = p$

**19.** *(d)* Let roots be $\alpha$ and $\beta$, then $\alpha + \beta = -1$

$\frac{1}{\alpha} + \frac{1}{\beta} = \frac{1}{6} \quad \Rightarrow \quad \frac{\beta + \alpha}{\alpha\beta} = \frac{1}{6}$

$\frac{-1}{\alpha\beta} = \frac{1}{6} \quad \Rightarrow \quad \alpha\beta = -6$

$\therefore$ Required equation is,

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$\Rightarrow \quad x^2 - (-1)x + (-6) = 0$

$\therefore \quad x^2 + x - 6 = 0$

**20.** *(c)* Let $\alpha$ and $\beta$ be the roots of the equation

$ax^2 + bx + c = 0$

$\therefore$ Sum of roots $(\alpha + \beta) = -\frac{b}{a}$

and product of roots $(\alpha\beta) = \frac{c}{a}$

By given condition,

$\alpha + \beta = \alpha^2 + \beta^2$

$\Rightarrow \quad \alpha + \beta = (\alpha + \beta)^2 - 2\alpha\beta$

$\Rightarrow \quad -\frac{b}{a} = \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right)$

$\Rightarrow \quad \frac{-b}{a} = \frac{b^2}{a^2} - \frac{2c}{a} \Rightarrow -ba = b^2 - 2ca$

$\Rightarrow \quad 2ac = b^2 + ab$

**21.** *(d)* Here, roots are $\alpha$ and $\alpha + 1$.

$\therefore \quad \alpha + (\alpha + 1) = l \qquad$ [sum of roots]

$\Rightarrow \quad 2\alpha = l - 1 \quad \Rightarrow \quad \alpha = \frac{l - 1}{2}$

Also, $\alpha(\alpha + 1) = m$ or $\alpha^2 + \alpha = m$

$\Rightarrow \quad \left(\frac{l - 1}{2}\right)^2 + \left(\frac{l - 1}{2}\right) = m$

$\Rightarrow \quad (l - 1)^2 + 2(l - 1) = 4m$

$\Rightarrow \quad l^2 - 1 = 4m \Rightarrow l^2 = 4m + 1$

**22.** *(b)* Here, $\alpha + \beta = (1 + a^2)$

and $\quad \alpha\beta = \frac{1}{2}(a^4 + a^2 + 1)$

$\because \quad \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= (1 + a^2)^2 - (a^4 + a^2 + 1)$

$= 1 + a^4 + 2a^2 - a^4 - a^2 - 1$

$\therefore \quad \alpha^2 + \beta^2 = a^2$

**23.** *(a)* Here, $\frac{1}{x + p} + \frac{1}{x + q} = \frac{1}{r}$

$\Rightarrow \quad r(x + p + x + q) = (x + p)(x + q)$

$\Rightarrow \quad x^2 + (p + q - 2r)x + pq - (p + q)r = 0$

Let roots be $\alpha$ and $(-\alpha)$, then

$\alpha + (-\alpha) = 0$

$\Rightarrow \quad -(p + q - 2r) = 0 \Rightarrow r = \frac{p + q}{2}$

So, product of roots $= pq - (p + q)r$

$= pq - (p + q) \cdot \frac{(p + q)}{2}$

$= pq - \frac{(p + q)^2}{2} = \frac{-1}{2}(p^2 + q^2)$

**24.** *(b)* Here, $\alpha + \beta = -\frac{2}{3}$ and $\alpha\beta = \frac{1}{3}$

$\Rightarrow \quad S = \frac{1 - \alpha}{1 + \alpha} + \frac{1 - \beta}{1 + \beta}$

$= \frac{(1 - \alpha)(1 + \beta) + (1 + \alpha)(1 - \beta)}{(1 + \alpha)(1 + \beta)}$

$= \frac{2 - 2\alpha\beta}{1 + (\alpha + \beta) + \alpha\beta} = \frac{2 - 2 \times \frac{1}{3}}{1 + (-2/3) + 1/3}$

$= \frac{4}{3} \times \frac{3}{2} = 2$

$P = \frac{1 - \alpha}{1 + \alpha} \times \frac{1 - \beta}{1 + \beta} = \frac{1 - (\alpha + \beta) + \alpha\beta}{1 + (\alpha + \beta) + \alpha\beta}$